Squamous Epithelium: It is made up of thin, **flat** cells with prominent nuclei. These cells have irregular boundaries and bind with neighbouring cells. The squamous epithelium is also known as **pavement membrane**, which form delicate lining of the buccal cavity, alveoli of lungs, proximal tubule of kidneys and covering of the skin and tongue. It protects the body from mechanical injury, drying and invasion of germs.

Cuboidal Epithelium: It is composed of single layer of **cubical** cells. The nucleus is round and lies in the centre. This tissue is present in the thyroid vesicles, salivary glands, sweat glands and exocrine pancreas. It is also found in the intestine and tubular part of the nephron (kidney tubules) as microvilli that increase the absorptive surface area. Their main function is secretion and absorption.

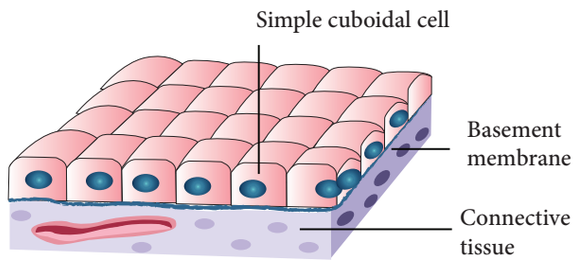


Figure 18.8 Cuboidal Epithelium

Columnar Epithelium: It is composed of a single layer of **slender**, **elongated** and **pillar** like cells. Their nuclei are located at the base. It is found lining the stomach, gall bladder, bile duct, small intestine, colon, oviducts and also forms the mucous membrane. They are mainly involved in secretion and absorption.

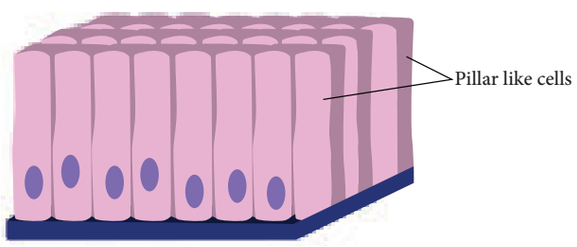


Figure 18.9 Columnar Epithelium

Ciliated Epithelium: Certain columnar cells bear numerous delicate hair like out growths called **cilia** and are called ciliated epithelium. Their function is to move particles or mucus in a specific direction over the epithelium. It is seen in the trachea of wind-pipe, bronchioles of respiratory tract, kidney tubules and fallopian tubes of oviducts.

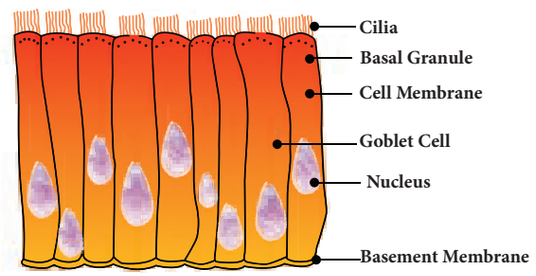


Figure 18.10 Ciliated Epithelium

Glandular Epithelium: Epithelial cells are often modified to form **specialized gland** cells which secrete chemical substances at the epithelial surface. This lines the gastric glands, pancreatic tubules and intestinal glands.

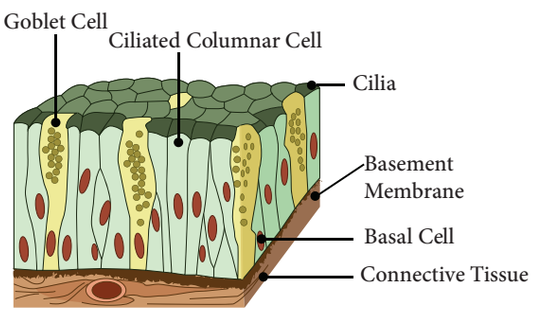


Figure 18.11 Glandular epithelium

B. Compound Epithelium

It consists of **more than one layer** of cells and gives a s**tratified appearance**. Hence, they are also known as stratified epithelium. The main function of this epithelium is to give protection to the underlying tissues against mechanical and chemical stress. They also cover the dry surface of the skin, the moist surface of the buccal cavity and pharynx.



Epithelial tissue in the skin functions as a water-proof membrane.

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